

# The Fractional Riccati–Müntz–Padé Theorem

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## Abstract

The classical theory of Padé approximation achieves its deepest results when the function being approximated is meromorphic outside a compact set of minimal logarithmic capacity—Stahl’s compact. The present paper extends this circle of ideas to solutions of fractional Riccati equations. Specifically, for the equation

$$D^\mu y = P(u) + Q(u)y + R(u)y^2 \quad (1)$$

with polynomial coefficients  $P, Q, R \in \mathbb{C}[u]$ , order  $\mu \in (0, 1]$ , and homogeneous Riemann–Liouville initial condition, the unique solution  $y(t, u)$  equals the limit, in logarithmic capacity, of its near-diagonal Müntz–Padé approximants of type  $[M - 1/M]$  in the variable  $z = t^\mu$ .

The argument turns on three interlocking observations. First, the solution admits an expansion  $y = \sum_k a(k, u) t^{(k+1)\mu}$  whose coefficients obey an explicit Müntz–Tau recurrence driven by the Fox–Wright weights  $\psi_k = \Gamma(k\mu+1)/\Gamma(k\mu+\mu+1)$ . Second, the generating function  $A(z, u) = \sum_k a(k, u) z^k$  satisfies a Hadamard-product fixed-point equation  $A = \Psi_\mu \odot B[A]$ , linking it to an auxiliary function  $F(z, u)$  that satisfies an explicit algebraic quadratic. Third, a Mellin–Barnes integral representation, valid under a Gevrey-1 bound established here from an explicit Cauchy-integral interpolation, transfers meromorphic continuation from the pole structure of  $\Psi_\mu$  to  $A$ , and a self-consistent fixed-point argument places  $K_A$  inside  $K_F$ .

The Padé denominators are identified as the orthogonal polynomials of the moment functional  $\mathcal{L}[x^k] = a(k, u)$ , computed by the Chebyshev–Wheeler algorithm, and convergence follows from a direct proof of Stahl’s theorem for the generic two-valued algebraic class containing  $F$ , given in the appendix.

## 1 Setup and Notation

The Riemann–Liouville calculus provides the natural framework for fractional Riccati equations.

**Definition 1** (Riemann–Liouville operators). *For  $\mu \in (0, 1]$  and  $f$  locally integrable on  $(0, \infty)$ ,*

$$I^\mu f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s) ds, \quad (2)$$

$$D^\mu f(t) := \frac{d}{dt} (I^{1-\mu} f)(t). \quad (3)$$

**Lemma 1** (Power rule). *For  $s > -1$ ,*

$$I^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+\mu+1)} t^{s+\mu}. \quad (4)$$

*Proof.* The substitution  $\sigma = t\tau$  in the defining integral (2) reduces the integral to the beta function

$$B(\mu, s+1) = \frac{\Gamma(\mu)\Gamma(s+1)}{\Gamma(s+\mu+1)}, \quad (5)$$

which gives (4). □

**Definition 2** (Fractional Riccati equation).

$$D^\mu y(t, u) = P(u) + Q(u) y(t, u) + R(u) y(t, u)^2, \quad I^{1-\mu} y(0, u) = 0. \quad (6)$$

**Lemma 2** (Volterra form). *Equation (6) is equivalent to*

$$y = I^\mu [P + Q y + R y^2]. \quad (7)$$

*Proof.* Application of  $I^\mu$  to (6), the composition formula [1, Theorem 2.4], and the homogeneous initial condition  $I^{1-\mu} y(0, u) = 0$  together kill the boundary term and yield (7).  $\square$

## 2 The Fox–Wright Weights

**Definition 3** (Fox–Wright weights and kernel).

$$\psi_k := \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)}, \quad k \geq 0, \quad (8)$$

$$\Psi_\mu(z) := \sum_{k=0}^{\infty} \psi_k z^k. \quad (9)$$

**Lemma 3** (Eigenvalue action). *For  $k \geq 0$ ,*

$$I^\mu t^{(k+1)\mu} = \psi_{k+1} t^{(k+2)\mu}. \quad (10)$$

*Proof.* Application of Lemma 1, equation (4), with  $s = (k+1)\mu$  gives (10).  $\square$

**Lemma 4** (Radius of convergence).  $\Psi_\mu$  has radius of convergence 1.

*Proof.* By Stirling,

$$\psi_k = \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)} \sim (k\mu)^{-\mu} \quad (k \rightarrow \infty). \quad (11)$$

Hence

$$\frac{1}{k} \log |\psi_k| = -\frac{\mu}{k} \log(k\mu) + o\left(\frac{1}{k}\right) \rightarrow 0, \quad (12)$$

so  $\limsup_{k \rightarrow \infty} |\psi_k|^{1/k} = 1$ , and the radius of convergence of  $\Psi_\mu$  is 1.  $\square$

## 3 The Müntz–Tau Series

**Theorem 1** (Müntz–Tau recurrence). *The unique solution of (7) admits the expansion*

$$y(t, u) = \sum_{k \geq 0} a(k, u) t^{(k+1)\mu} \quad (13)$$

on  $[0, T_0]$  for some  $T_0 > 0$ , with

$$a(0, u) = P(u) \psi_0, \quad (14)$$

$$a(k, u) = \psi_k \left( Q(u) a(k-1, u) + R(u) \sum_{j+\ell=k-1} a(j, u) a(\ell, u) \right), \quad k \geq 1. \quad (15)$$

*Proof.* The map

$$\mathcal{T}[y] := I^\mu(P + Qy + Ry^2) \quad (16)$$

is a contraction on the ball  $\{y \in C[0, T_0] : \|y\|_\infty \leq 2|P(u)|T_0^\mu/\Gamma(\mu+1)\}$  whenever

$$T_0^\mu < \frac{\Gamma(\mu+1)}{|Q(u)| + 2|R(u)|\|y\|_\infty}, \quad (17)$$

making the local existence radius explicit and  $u$ -dependent. Substitution of the Müntz ansatz (13) and application of Lemma 3 match coefficients of  $t^{(k+1)\mu}$ :  $k=0$  gives (14);  $k \geq 1$  gives (15).  $\square$

## 4 The Algebraic Quadratic via Generating Functions

**Conventions.** Coefficient extraction

$$[z^k]b := b_k, \quad (18)$$

Cauchy product

$$[z^k](F \cdot G) = \sum_j f_j g_{k-j}, \quad (19)$$

Hadamard product

$$[z^k](F \odot G) = f_k g_k. \quad (20)$$

**Definition 4** (Auxiliary generating function). Set  $f(0, u) := P\psi_0$  and, for  $k \geq 1$ ,

$$f(k, u) := Qf(k-1, u) + R \sum_{j+\ell=k-1} f(j, u)f(\ell, u) + P\psi_k. \quad (21)$$

Define

$$F(z, u) := \sum_{k \geq 0} f(k, u) z^k. \quad (22)$$

**Remark 1.** In (15) the bracket is premultiplied by  $\psi_k$ ; in (21) the term  $P\psi_k$  is a separate source. Hence  $F \neq A$  and neither  $\Psi_\mu \cdot A$  nor  $\Psi_\mu \odot A$  equals  $F$ . The bridge is the Hadamard fixed point of Section 7. Comparison of first-order coefficients gives

$$a(1, u) = \psi_1(QP\psi_0 + R(P\psi_0)^2), \quad (23)$$

$$f(1, u) = QP\psi_0 + R(P\psi_0)^2 + P\psi_1, \quad (24)$$

which differ as elements of  $\mathbb{C}[u]$  for generic  $P, Q, R, \mu$ .

**Theorem 2** (Algebraic quadratic for  $F$ ).  $F$  satisfies

$$zRF^2 - (1 - zQ)F + P\Psi_\mu = 0, \quad (25)$$

with the closed form

$$F(z, u) = \frac{(1 - zQ) - \sqrt{(1 - zQ)^2 - 4zPR\Psi_\mu(z)}}{2zR}. \quad (26)$$

*Proof.* Coefficient-by-coefficient,

$$G := zRF^2 - (1 - zQ)F + P\Psi_\mu \quad (27)$$

satisfies  $[z^0]G = 0$  and  $[z^k]G = 0$  for  $k \geq 1$  by (21). Expansion of the discriminant,

$$\sqrt{\Delta} = 1 - z(Q + 2PR\psi_0) + O(z^2), \quad (28)$$

shows the minus branch of (26) has constant term  $P\psi_0 = f(0, u)$ , the unique formal power-series root since

$$\partial_F[zRF^2 - (1 - zQ)F + P\Psi_\mu]|_{z=0} = -1. \quad (29)$$

$\square$

## 5 The Chebyshev–Wheeler Algorithm

### 5.1 Points of growth and positive-definiteness

**Definition 5.** For  $\lambda$  non-decreasing right-continuous,  $x_0$  is a point of growth if  $\lambda(x_0 + \varepsilon) - \lambda(x_0 - \varepsilon) > 0$  for all  $\varepsilon > 0$ . Set

$$G(\lambda) := \text{supp}(d\lambda), \quad N_\lambda := |G(\lambda)|. \quad (30)$$

**Lemma 5.** The form

$$\langle f, g \rangle_\lambda := \int f g d\lambda \quad (31)$$

is positive-definite on  $\mathbb{R}[x]$  iff  $N_\lambda = \infty$ .

*Proof.* ( $\Leftarrow$ ) A degree- $d$  polynomial vanishes at  $\leq d$  points, so  $f^2$  is positive on some point of growth. ( $\Rightarrow$ ) If  $G(\lambda) = \{x_1, \dots, x_M\}$ , then

$$f(x) := \prod_{j=1}^M (x - x_j) \quad (32)$$

vanishes on  $G(\lambda)$ , giving  $\int f^2 d\lambda = 0$ .  $\square$

### 5.2 Wheeler's algorithm

**Auxiliary array.** Initial data and recurrence:

$$\sigma(-1, \ell) = 0, \quad (33)$$

$$\sigma(0, \ell) = \mu(\ell), \quad (34)$$

$$\sigma(k, \ell) = \sigma(k-1, \ell+1) - \alpha(k-1)\sigma(k-1, \ell) - \beta(k-1)\sigma(k-2, \ell). \quad (35)$$

**Coefficients.**

$$\alpha(0) = \mu(1)/\mu(0), \quad \beta(0) = \mu(0), \quad (36)$$

$$\alpha(k) = \frac{\sigma(k, k+1)}{\sigma(k, k)} - \frac{\sigma(k-1, k)}{\sigma(k-1, k-1)}, \quad (37)$$

$$\beta(k) = \frac{\sigma(k, k)}{\sigma(k-1, k-1)}. \quad (38)$$

**Orthogonal polynomials.**

$$\pi_{-1} = 0, \quad \pi_0 = 1, \quad (39)$$

$$\pi_{k+1} = (x - \alpha(k))\pi_k - \beta(k)\pi_{k-1}. \quad (40)$$

**Theorem 3** (Wheeler).  $\sigma(k, k) > 0$  and

$$\int \pi_j \pi_k d\lambda = \delta_{jk} \prod_{i=0}^k \beta(i). \quad (41)$$

*Proof.* Set

$$\tilde{\sigma}(k, \ell) := \int \pi_k x^\ell d\lambda. \quad (42)$$

This matches  $\sigma$  in initial data and recurrence, so they coincide. Induction shows  $\pi_k \perp \{1, \dots, x^{k-1}\}$ , giving  $\sigma(k, \ell) = 0$  for  $\ell < k$  and  $\sigma(k, k) = \langle \pi_k, \pi_k \rangle_\lambda > 0$  by Lemma 5. The norm formula follows from

$$\langle \pi_k, \pi_k \rangle = \beta(k) \langle \pi_{k-1}, \pi_{k-1} \rangle. \quad (43)$$

$\square$

### 5.3 Symbolic transfer

**Definition 6.**  $\{a(k, u)\}$  is quasi-definite if the Hankel determinants

$$H_n(u) := \det(a(i+j, u))_{0 \leq i, j < n} \quad (44)$$

satisfy  $H_n(u) \neq 0$  for all  $n$ .

**Theorem 4** (Symbolic Wheeler). *Under quasi-definiteness,  $\sigma(k, k, u) \neq 0$ , and the orthogonal polynomials  $P_k$  of the functional  $\mathcal{L}[x^\ell] = a(\ell, u)$  satisfy*

$$\mathcal{L}[P_j P_k] = \delta_{jk} \prod_{i=0}^k \beta_i(u) \quad \text{in } \mathbb{C}(u). \quad (45)$$

*Proof.* The Hankel ratio gives

$$\sigma(k, k, u) = \frac{H_{k+1}(u)}{H_k(u)} \neq 0. \quad (46)$$

The identity

$$\sigma(k, \ell, u) = \mathcal{L}[x^\ell P_k] \quad (47)$$

holds by induction, and  $\mathcal{L}[P_k^2] = \sigma(k, k, u)$  telescopes to (45).  $\square$

## 6 Padé Denominators via Gragg's Theorem

**Padé convention.** The  $M$ -th approximant has  $\deg N_M \leq M-1$ ,  $\deg D_M \leq M$ , and is therefore of type  $[M-1/M]$ . Its denominator sequence  $\{D_M\}$  is identical to that of the diagonal  $[M/M]$  table; these approximants are *diagonal*, following Stahl.

**Theorem 5** (Gragg, symbolic form). *There is a unique pair  $(N_M, D_M)$  with  $\deg N_M \leq M-1$ ,  $\deg D_M \leq M$ ,  $D_M(0, u) = 1$ , and*

$$D_M A - N_M = O(z^{2M}), \quad (48)$$

*satisfying*

$$D_M(z, u) = z^M P_M(z^{-1}, u). \quad (49)$$

*Proof.* The  $M$  matching conditions

$$[z^j](D_M A) = 0, \quad j = M, \dots, 2M-1, \quad (50)$$

become, with  $Q_M(x, u) := x^M D_M(x^{-1}, u)$ , the orthogonality

$$\mathcal{L}[x^i Q_M] = 0, \quad i = 0, \dots, M-1. \quad (51)$$

The monic  $Q_M$  of degree  $\leq M$  matching these conditions is unique by Theorem 4:  $Q_M = P_M$ . Since  $P_M$  is monic,  $D_M(0, u) = 1$ .  $\square$

## 7 Meromorphic Continuation of $A$

This section establishes that  $A$  is meromorphic on  $\tilde{X} \setminus K_A(u)$ , the universal cover of  $\mathbb{C} \setminus \{0\}$  minus a compact  $K_A(u) \subseteq K_F(u)$  of minimal logarithmic capacity.

## 7.1 Hadamard fixed-point equation

**Lemma 6.** *With*

$$B[A] := P + zQA + zRA^2, \quad (52)$$

*the identity*

$$A = \Psi_\mu \odot B[A] \quad (53)$$

*holds.*

*Proof.*  $[z^0]B[A] = P$  gives  $\psi_0 P = a(0)$ . For  $k \geq 1$ ,

$$[z^k]B[A] = Q a(k-1) + R \sum_{j+\ell=k-1} a(j) a(\ell), \quad (54)$$

and multiplication of (54) by  $\psi_k$  recovers  $a(k)$  by (15).  $\square$

## 7.2 Mellin transform of $\Psi_\mu$

**Lemma 7** (Mellin transform). *For  $0 < \Re s < 1$ ,*

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \Gamma(1-s) \frac{\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)} =: M_{\Psi_\mu}(s). \quad (55)$$

$M_{\Psi_\mu}$  extends meromorphically to  $\mathbb{C}$  with poles at  $\mathbb{Z}_{\leq 0}$ ,  $\mathbb{Z}_{\geq 1}$ , and  $(1+m)/\mu$  for  $m \geq 0$ . A pole at  $(1+m)/\mu$  is cancelled by a denominator pole at  $1+(1+n)/\mu$  iff  $m-n=\mu$ ; for  $\mu \in (0,1]$  this requires  $\mu=1$ . For  $\mu \in (0,1)$  the lattice  $\{(1+m)/\mu\}$  survives uncanceled. At  $\mu=1$  the collapse is explicit:

$$M_{\Psi_1}(s) = \Gamma(s) \Gamma(1-s) \frac{\Gamma(1-s)}{\Gamma(2-s)} = \frac{\Gamma(s) \Gamma(1-s)}{1-s}, \quad (56)$$

whose order-two pole at  $s=1$  is precisely what produces the logarithmic kernel  $\Psi_1(z) = -\log(1-z)/z$ : a simple Mellin pole would give a power law, the double pole gives the log.

*Proof.* Write

$$\Psi_\mu(-x) = \sum_{k \geq 0} \frac{(-x)^k \varphi(k)}{k!}, \quad \varphi(k) := k! \psi_k. \quad (57)$$

The interpolation

$$\varphi(s) = \frac{\Gamma(s+1) \Gamma(s\mu+1)}{\Gamma(s\mu+\mu+1)} \quad (58)$$

is entire. Stirling on a vertical line  $\Re s = \sigma$  gives

$$|\varphi(\sigma + i\tau)| \sim C(\sigma, \mu) |\tau|^{\sigma+\mu\sigma-\mu+1/2} e^{-(\pi/2)|\tau|} \quad (|\tau| \rightarrow \infty), \quad (59)$$

where  $C(\sigma, \mu) > 0$  depends only on  $\sigma$  and  $\mu \in (0,1]$ . The exponential rate  $\pi/2 < \pi$  satisfies Hardy's growth condition [8, Lecture XI, §§11.2, 11.10–11.13], and Ramanujan's master theorem gives

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \varphi(-s). \quad (60)$$

The cancellation analysis is recorded in the statement.  $\square$

### 7.3 Mellin–Barnes realisation

**Lemma 8.** *Let*

$$G(z, u) = \sum_{k \geq 0} g_k(u) z^k \quad (61)$$

*be holomorphic in  $|z| < \rho$ , with Ramanujan-normalised entire interpolation*

$$\widehat{G}_R(s, u) := \frac{\widehat{G}(s, u)}{\Gamma(s+1)}, \quad (62)$$

*where  $\widehat{G}$  is the Cauchy interpolation of (82) below, satisfying  $\widehat{G}_R(k, u) = g_k(u)$  at integers and the Gevrey-1 bound*

$$|\widehat{G}(\sigma + i\tau, u)| \leq K'_G C_G^{|\sigma|+1} |\Gamma(\sigma + i\tau + 1)|. \quad (63)$$

*Fix  $c \in (0, 1)$  avoiding poles of  $M_{\Psi_\mu}$ . For  $|\arg(-z)| < \pi/2$  and  $|z| < \min(1, \rho)$ ,*

$$(\Psi_\mu \odot G)(z, u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s) \widehat{G}_R(-s, u) (-z)^{-s} ds. \quad (64)$$

*The integrand decays as*

$$e^{-(\pi/2)|\tau|} |\tau|^{C_1+C_2}, \quad (65)$$

*so the integral converges absolutely on the sector and continues meromorphically to  $\widetilde{X}$  minus the projected pole set of  $M_{\Psi_\mu} \widehat{G}_R$ .*

*Proof.* Closing the contour to the left, the simple pole of  $\Gamma(s)$  at  $s = -k$  has residue  $(-1)^k/k!$ . Split

$$M_{\Psi_\mu}(s) = \Gamma(s) \cdot H(s), \quad H(s) := \frac{\Gamma(1-s) \Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)}, \quad (66)$$

so that  $\Gamma(s)$  is the only factor producing a pole at  $s = -k$ . At  $s = -k$ ,

$$H(-k) = \Gamma(1+k) \frac{\Gamma(1+k\mu)}{\Gamma(\mu+1+k\mu)} = k! \psi_k, \quad (67)$$

$$\widehat{G}_R(-s, u)|_{s=-k} = \widehat{G}_R(k, u) = g_k(u), \quad (68)$$

$$(-z)^{-s}|_{s=-k} = (-1)^k z^k. \quad (69)$$

Therefore

$$\operatorname{Res}_{s=-k} [M_{\Psi_\mu}(s) \widehat{G}_R(-s, u) (-z)^{-s}] = \frac{(-1)^k}{k!} \cdot k! \psi_k \cdot g_k(u) \cdot (-1)^k z^k = \psi_k g_k(u) z^k, \quad (70)$$

using  $(-1)^k(-1)^k = 1$ . Summation over  $k \geq 0$  recovers

$$\sum_{k \geq 0} \psi_k g_k z^k = \Psi_\mu \odot G. \quad (71)$$

Continuation to  $\widetilde{X}$  is by the identity principle along chains of overlapping disks.  $\square$

## 7.4 Stahl compact for $F$ and for $A$

**Lemma 9.** *There is a unique compact  $K_F(u) \supseteq K_\Psi$  of minimal capacity outside which  $\sqrt{\Delta_F}$ , hence  $F$ , is single-valued meromorphic on  $\tilde{X}$ . Here*

$$\Delta_F = (1 - zQ)^2 - 4zPR\Psi_\mu, \quad (72)$$

and  $K_\Psi$  is the projected pole set of  $M_{\Psi_\mu}$ .

*Proof.* By Lemma 7,  $\Delta_F$  is meromorphic on  $\tilde{X} \setminus K_\Psi$ . Theorem 9 of the appendix produces the unique minimal-capacity compact  $K_F(u) \supseteq K_\Psi$  outside which  $\sqrt{\Delta_F}$  is single-valued.  $\square$

**Theorem 6** (Meromorphic continuation of  $A$ ). *Under quasi-definiteness,  $A$  is meromorphic on  $\tilde{X} \setminus K_A(u)$  with  $K_A(u) \subseteq K_F(u)$ .*

*Proof. Step 1: Gevrey-1 bound on  $\{a(k, u)\}$  and  $\{g_k(u)\}$ .* Fix  $u$  in a compact subset of parameter space. Since  $\psi_k$  is decreasing in  $k$  for fixed  $\mu \in (0, 1]$ , the supremum is attained at  $k = 0$ :

$$C_0 := \sup_{k \geq 0} |\psi_k| = \psi_0 = \frac{1}{\Gamma(\mu + 1)} < \infty \quad (\text{by Lemma 4}). \quad (73)$$

Set

$$M_P := |P(u)|, \quad M_Q := |Q(u)|, \quad M_R := |R(u)|, \quad (74)$$

and define the (local) Gevrey constants

$$K_G := C_0 \max(M_P, 1), \quad C_G := C_0 (M_Q + 2K_G M_R) + 1. \quad (75)$$

The bound to be established is

$$|a(k, u)| \leq K_G C_G^k k! \quad \text{for all } k \geq 0. \quad (76)$$

The base case  $k = 0$  uses  $|a(0, u)| = |P\psi_0| \leq C_0 M_P \leq K_G \leq K_G C_G^0 0!$ . Inductively, assuming (76) through  $k - 1$ ,

$$|a(k, u)| \leq C_0 (M_Q |a(k - 1)| + M_R \sum_{j+\ell=k-1} |a(j)| |a(\ell)|). \quad (77)$$

The convolution sum is bounded using

$$\sum_{j+\ell=n} j! \ell! \leq (n + 1)!, \quad (78)$$

giving  $\sum_{j+\ell=k-1} K_G^2 C_G^{k-1} j! \ell! \leq K_G^2 C_G^{k-1} k!$ . Hence

$$|a(k)| \leq C_0 M_Q K_G C_G^{k-1} (k - 1)! + C_0 M_R K_G^2 C_G^{k-1} k! \leq K_G C_G^k k! \cdot \frac{C_0 (M_Q + K_G M_R)}{C_G} \leq K_G C_G^k k!, \quad (79)$$

since  $C_0 (M_Q + K_G M_R) / C_G < 1$  by (75). The bound for  $\{g_k(u)\}$  follows from

$$g_k = Q a(k - 1) + R \sum_{j+\ell=k-1} a(j) a(\ell) + P[k = 0], \quad (80)$$

yielding

$$|g_k(u)| \leq K_G'' C_G^{k+1} k!, \quad K_G'' := K_G (M_Q + K_G M_R) / C_G^2. \quad (81)$$

**Step 2: Cauchy-integral interpolation.** Fix  $0 < r < \rho_0$  where  $G$  is holomorphic. Define

$$\hat{G}(s, u) := \Gamma(s + 1) \cdot \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz, \quad \hat{G}_R(s, u) := \frac{\hat{G}(s, u)}{\Gamma(s + 1)}. \quad (82)$$

At integer points,

$$\widehat{G}(k, u) = k! g_k(u), \quad \widehat{G}_R(k, u) = g_k(u), \quad (83)$$

the latter being the Ramanujan normalisation used in Lemma 8. The Cauchy-kernel factor

$$\kappa(s, u) := \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz \quad (84)$$

is entire in  $s$  of exponential type at most  $\log(1/r)$  along the imaginary axis. Multiplication by  $\Gamma(s+1)$  does not alter this type. Carlson's theorem requires

$$\log(1/r) < \pi \iff r > e^{-\pi}. \quad (85)$$

Since  $G$  is holomorphic on  $|z| < \rho_0$ ,  $\kappa(s, u)$  is independent of  $r$  in  $(0, \rho_0)$ , so  $r$  may be enlarged to any value in  $(e^{-\pi}, \rho_0)$ , which is nonempty provided  $\rho_0 > e^{-\pi}$ . If  $\rho_0 \leq e^{-\pi}$ , fix any  $r_0 \in (0, \rho_0)$  and replace  $z$  by  $\lambda z$  for a scalar  $\lambda$  with

$$\lambda r_0 \in (e^{-\pi}, 1); \quad (86)$$

this rescaling reduces to the previous case and is undone after the Mellin–Barnes representation by the inverse scaling. Carlson's theorem then gives uniqueness of  $\widehat{G}$  as the entire interpolation of  $\{k! g_k(u)\}_{k \geq 0}$  of exponential type  $< \pi$ . The Cauchy integral satisfies

$$|\kappa(s, u)| \leq \frac{M(r, u)}{r^{\Re s}}, \quad M(r, u) := \sup_{|z|=r} |G(z, u)|, \quad (87)$$

which gives

$$|\widehat{G}(\sigma + i\tau, u)| \leq \frac{M(r, u) |\Gamma(\sigma + i\tau + 1)|}{r^\sigma}. \quad (88)$$

Combination of (88) with the integer-point Gevrey bound (81) and the Phragmén–Lindelöf principle in each vertical strip (applied to  $\widehat{G}(s, u)/[K'_G C_G^{|\sigma|+1} \Gamma(s+1)]$ , which is bounded on the imaginary axis and on  $\Re s = N$  for each  $N \in \mathbb{Z}_{\geq 0}$  by the Gevrey bound and Stirling, and of finite order in the strip) yields (63) with

$$K'_G := M(r, u)/r, \quad (89)$$

and  $C_G$  as in (75) (after enlarging  $C_G$  to absorb  $1/r$ ).

**Step 3: Mellin–Barnes representation.** By Lemma 6,

$$A = \Psi_\mu \odot G, \quad G = B[A]. \quad (90)$$

Lemma 8 applies and yields, for  $|\arg(-z)| < \pi/2$  and  $|z| < \min(1, \rho_0)$ , the integral representation (64), holomorphic on the open sector. By the identity principle,  $A$  extends meromorphically to

$$\widetilde{X} \setminus (K_\Psi \cup S_G), \quad (91)$$

where  $S_G$  is the singularity locus of  $\widehat{G}_R$ .

**Step 4: Self-consistent fixed point.** For  $K \supseteq K_\Psi$  compact, let  $\mathcal{M}(K)$  be the space of  $\mathbb{C}[u][[z]]$ -formal series whose unique meromorphic continuation lives on  $\widetilde{X} \setminus K$ . The map

$$\Phi : A \mapsto \Psi_\mu \odot B[A] \quad (92)$$

sends  $\mathcal{M}(K)$  to itself:  $B[A] = P + zQA + zRA^2$  is polynomial in  $A$ , hence in  $\mathcal{M}(K)$ , and the Mellin–Barnes integral adds only the poles of  $M_{\Psi_\mu}$ , which project to  $K_\Psi \subseteq K$ . The unique formal-power-series fixed point of  $\Phi$  is  $A$  (Theorem 1).  $F$  is the unique formal-power-series root of (25) and lies in  $\mathcal{M}(K_F)$  (Lemma 9). Subtraction of the  $F$ -equation from the  $A$ -equation and order-by-order reading of the difference shows  $A - F$  has at worst the same singular set as  $F$  (Lemma 13). Therefore  $A \in \mathcal{M}(K_F(u))$ , and the minimal compact  $K_A(u) \subseteq K_F(u)$  exists by Theorem 9.  $\square$

## 8 The Fractional Riccati–Müntz–Padé Theorem

**Theorem 7** (Fractional Riccati–Müntz–Padé theorem). *Under quasi-definiteness, the Müntz–Padé approximant*

$$y_M(t, u) := t^\mu \frac{N_M(t^\mu, u)}{D_M(t^\mu, u)} \quad (93)$$

*satisfies*

$$y(t, u) = \lim_{M \rightarrow \infty} y_M(t, u) \quad (94)$$

*in logarithmic capacity on compact subsets of the  $t$ -universal-cover preimage of  $\tilde{X} \setminus K_A(u)$ .*

*Proof.* By Theorem 1,

$$y(t, u) = t^\mu A(t^\mu, u); \quad (95)$$

$N_M/D_M$  is the  $[M - 1/M]$  Padé approximant of  $A$  by Theorem 5.

*Equivalence of denominator sequences.*  $D_M$  is determined by (51). The  $[M/M]$  approximant adds one matching condition that fixes the leading coefficient of  $N_M$ , not the denominator. Hence  $\{D_M\}_{M \geq 1}$  is identical for both types and Stahl’s theorem applies to either. By quasi-definiteness the Padé table of  $A(\cdot, u)$  is normal: all Hankel determinants  $H_n(u) \neq 0$ , and the  $[M - 1/M]$  and  $[M/M]$  entries share the denominator  $\pi_M(\cdot, u)$  (the Gragg connection, Theorem 5).

*Convergence in capacity.* By Theorem 6,  $A$  is meromorphic on  $\tilde{X} \setminus K_A(u)$ . Theorem 10 of the appendix gives

$$\frac{N_M}{D_M} \longrightarrow A \quad \text{in capacity on compacta of } \tilde{X} \setminus K_A(u). \quad (96)$$

*Transfer to the  $t$ -plane.* Fix the principal-branch sector

$$\Sigma := \{t \in \tilde{\mathbb{C}}^* : \arg t \in (-\pi, \pi]\}, \quad (97)$$

on which  $z = t^\mu$  is a biholomorphism onto its image. The map  $t \mapsto t^\mu$  is a  $C^1$  diffeomorphism on  $\Sigma \setminus \{0\}$  with bounded distortion on compacta, and the classical capacity-transformation identity (cf. Ransford [9, Thm. 5.2.5]) gives

$$\text{cap}(\{t \in \Sigma : t^\mu \in K\}) = \text{cap}(K)^{1/\mu} \quad (98)$$

for every compact  $K$  in the image. On the full universal cover of  $\mathbb{C} \setminus \{0\}$ , the preimage decomposes into countably many such sectorial sheets, and convergence in capacity on each sheet transfers from the  $z$ -plane to the  $t$ -plane sheetwise. The statement of the theorem refers to this sheetwise transfer.  $\square$

**Lemma 10** (No real poles). *In the constant-coefficient case  $P(u), Q(u), R(u) \in \mathbb{C}$  with  $R(u) \neq 0$ , the generating function  $A(z, u)$  has no poles on  $(0, \infty)$ .*

*Proof.* Each pole  $z^*$  of  $A(\cdot, u)$  is simple, with residue

$$\text{Res}(A(\cdot, u), z^*) = -\frac{(z^*)^{(\mu-1)/\mu}}{R(u)\mu}. \quad (99)$$

For  $z^* \in (0, \infty)$  this residue is nonzero; but the Volterra fixed point is holomorphic on every compact real interval on which it converges, a contradiction. Hence  $A(\cdot, u)$  has no poles on  $(0, \infty)$ .  $\square$

**Corollary 1** (Uniform convergence on  $\mathbb{R}_+$ ). *In the constant-coefficient case, for every compact  $[0, T] \subset \mathbb{R}_+$ ,*

$$\sup_{t \in [0, T]} |y(t) - y_M(t)| \longrightarrow 0 \quad \text{as } M \rightarrow \infty. \quad (100)$$

*Proof.* By Lemma 10,  $\delta := \text{dist}([0, T], \mathcal{P}_A) > 0$ , where  $\mathcal{P}_A$  denotes the pole set of  $A(\cdot, u)$ . On  $U_\delta := \{z \in \mathbb{C} : \text{dist}(z, [0, T]) < \delta/2\}$ ,  $A(\cdot, u)$  is holomorphic and bounded. Convergence in capacity on  $U_\delta$  (Theorem 10) upgrades to uniform convergence on  $[0, T]$ : the capacity-zero exceptional sets cannot meet  $[0, T]$  for all sufficiently large  $M$ , and a normal-families bound finishes the argument. The substitution  $z = t^\mu$  transfers uniform convergence from  $[0, T^\mu]$  to  $[0, T]$ .  $\square$

## 9 Exact Option Pricing from the Müntz–Padé Characteristic Function

Given the characteristic function in the form produced by the Müntz–Padé expansion, the European option price is computed *exactly*: by a finite sum of residues, with no quadrature, no contour truncation, no damping parameter, and no fast Fourier transform. We prove the pricing identity (Theorem 8), its error bound (Proposition 1), and its optimality (Remark 2).

**Standing setup.** Fix maturity  $T$  and rate  $r = 0$ ; the general case differs only by the discount factor  $e^{-rT}$  and the forward shift  $k \mapsto k - rT$ . Let  $X_T = \log(S_T/S_0)$  and write the order- $M$  characteristic function as

$$\varphi_M(v) = \exp(\Phi_M(v)), \quad \Phi_M(v) = \sum_{j=1}^M \frac{c_j}{v - \eta_j}, \quad (101)$$

the partial fraction of the rational cumulant exponent  $\Phi_M$  in the Fourier variable  $v$ , with simple poles  $\{\eta_j\}_{j=1}^M$  and residues  $c_j$  read off the orthogonal-polynomial recurrence (Theorem 5). The normalisations

$$\varphi_M(0) = 1, \quad \varphi_M(-i) = \mathbb{E}[e^{X_T}] = 1 \quad (102)$$

hold *exactly for every*  $M$ , because the Riccati source  $P(v) = \frac{1}{2}(-v^2 - iv)$  vanishes at  $v = 0$  and  $v = -i$ , so every  $a(k, 0) = a(k, -i) = 0$  and  $\Phi_M(0) = \Phi_M(-i) = 0$ . By Hermitian symmetry  $\varphi_M(-\bar{v}) = \overline{\varphi_M(v)}$  and the no-real-poles property (Lemma 10, transported to the  $v$ -plane through the analyticity strip of the exact  $\Phi$ ), the poles  $\eta_j$  lie off  $\mathbb{R}$  and occur in conjugate pairs.

**Lemma 11** (Payoff transform). *Let  $k = \log(K/S_0)$  and  $w(x) = (S_0 e^x - K)^+$ . For  $\text{Im } v < -1$ ,*

$$\int_{\mathbb{R}} e^{-ivx} w(x) dx = -\frac{K e^{-ivk}}{v(v+i)}. \quad (103)$$

*Proof.* For  $\text{Im } v < -1$ ,  $\int_k^\infty e^{-ivx} S_0 e^x dx = -S_0 e^{(1-iv)k} / (1-iv)$ , and for  $\text{Im } v < 0$ ,  $\int_k^\infty e^{-ivx} (-K) dx = -K e^{-ivk} / (iv)$ . With  $S_0 e^k = K$  and  $\frac{1}{1-iv} + \frac{1}{iv} = \frac{1}{v(v+i)}$ , the two combine to (103).  $\square$

**Theorem 8** (Exact European call price). *With  $k = \log(K/S_0)$  and any  $\alpha \in (1, p^*)$ , where  $-ip^*$  is the  $v$ -pole of  $\varphi_M$  nearest the strip  $\{\text{Im } v = -\alpha\}$  from below,*

$$C_M = -\frac{K}{2\pi} \int_{\text{Im } v = -\alpha} \frac{e^{-ivk} \varphi_M(v)}{v(v+i)} dv. \quad (104)$$

*Closing the contour gives the finite, exact evaluation*

$$C_M = \begin{cases} (S_0 - K) - K \operatorname{Re} \left[ i \sum_{\text{Im } \eta_j > -\alpha} \mathcal{R}_j \right], & k \leq 0, \\ K \operatorname{Re} \left[ i \sum_{\text{Im } \eta_j < -\alpha} \mathcal{R}_j \right], & k > 0, \end{cases} \quad (105)$$

where, writing  $\varphi_M(v) = \exp(c_j/(v - \eta_j)) H_j(v)$  with  $H_j$  holomorphic and nonvanishing at  $\eta_j$  and  $G_j(v) := e^{-ivk} H_j(v)/(v(v+i))$ ,

$$\mathcal{R}_j = \operatorname{Res}_{v=\eta_j} \frac{e^{-ivk} \varphi_M(v)}{v(v+i)} = \sum_{n \geq 1} \frac{c_j^n G_j^{(n-1)}(\eta_j)}{n! (n-1)!} \quad (106)$$

*is an absolutely convergent  ${}_0F_1$ -type series.*

*Proof. Parseval.* With density  $f$  of  $X_T$  and  $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ivx} \varphi_M(v) dv$ ,

$$C_M = \mathbb{E}[w(X_T)] = \int_{\mathbb{R}} w(x) f(x) dx = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi_M(v) \int_{\mathbb{R}} w(x) e^{-ivx} dx dv.$$

The inner integral converges only for  $\text{Im } v < -1$ , so shift the  $v$ -contour to  $\text{Im } v = -\alpha$  ( $\alpha \in (1, p^*)$ ), legitimate since  $\varphi_M$  is holomorphic and  $O(1)$  on the strip  $-p^* < \text{Im } v < 0$  (no real poles, conjugate-pair poles bounded away from the strip). Lemma 11 gives (104).

*Contour closure.* The integrand has simple kernel poles at  $v = 0, -i$  and essential singularities at the  $\eta_j$ . For  $k \leq 0$  the factor  $e^{-ivk}$  decays as  $\text{Im } v \rightarrow +\infty$ , so close upward; the enclosed poles are  $v = 0, v = -i$  (both above  $-\alpha$  since  $\alpha > 1$ ), and the  $\eta_j$  with  $\text{Im } \eta_j > -\alpha$ . By residues  $C_M = -Ki \sum_{\text{encl}} \text{Res}$ .

*Kernel poles give the intrinsic.* Using (102),  $\text{Res}_{v=0} = \varphi_M(0)/i = -i$  and  $\text{Res}_{v=-i} = e^{-k} \varphi_M(-i)/(-i) = i e^{-k} = i S_0/K$ . Hence  $-Ki(-i) + (-Ki)(i S_0/K) = -K + S_0 = S_0 - K$ , the exact intrinsic.

*Model poles give the series.* At  $\eta_j$ ,  $\varphi_M(v) = \exp(c_j/(v - \eta_j)) H_j(v)$  with  $H_j$  holomorphic there. Multiplying the Laurent series  $\exp(c_j/(v - \eta_j)) = \sum_{n \geq 0} c_j^n / (n! (v - \eta_j)^n)$  by the Taylor series of  $G_j$  and extracting the  $(v - \eta_j)^{-1}$  coefficient gives (106). The bound  $|c_j^n G_j^{(n-1)}(\eta_j) / (n! (n-1)!)| \leq \|G_j\|_B |c_j|^n / (n! (n-1)! \rho_j^{n-1})$ , with  $\rho_j = \min_{i \neq j} |\eta_i - \eta_j|$ , is summable and super-exponentially small, so (106) converges absolutely. Taking the real part (the price is real) yields (105); the  $k > 0$  branch closes downward, enclosing neither kernel pole, so its intrinsic vanishes.  $\square$

**Proposition 1** (Pricing error). *On compact strike-maturity sets there is  $r(K, T) \in (0, 1)$  and  $C'(K, T)$  with  $|C - C_M| \leq C'(K, T) r(K, T)^M$ , where  $C = \lim_{M \rightarrow \infty} C_M$  is the exact price.*

*Proof.*  $|\varphi - \varphi_M| \leq e^{\max(|\Phi|, |\Phi_M|)} |\Phi - \Phi_M|$ , and Theorem 7 bounds  $|\Phi - \Phi_M|$  in capacity by  $C_0 r^M$  on the integration strip; integrating against the fixed kernel  $1/(v(v+i))$  over  $\{\text{Im } v = -\alpha\}$  preserves the geometric rate.  $\square$

**Remark 2** (Optimality). *The evaluation (105) is optimal for this representation:*

1. Single error source. *The only approximation is the Padé order  $M$ , controlling  $\Phi$  at the capacity rate (Theorem 7); each per-pole series (106) converges super-exponentially in its truncation order  $L$ . A quadrature inversion (Carr–Madan, COS, Lewis) instead carries two error sources — the Padé order and a discretisation/contour parameter — and converges only geometrically.*
2. Cost. *The orthogonal-polynomial recurrence and the poles/residues cost  $O(M^2)$  once, reused across all strikes and maturities; each strike then costs  $O(L^2 M)$  to assemble (105).*
3. Conditioning. *The poles are zeros of the orthogonal polynomials of a Stieltjes/Markov functional, hence simple and separated with no spurious (Froissart) doublets (complete monotonicity  $\Rightarrow$  positive measure  $\Rightarrow$  Markov; Lemma 10, Theorem 10). Thus  $\rho_j$  in (106) is bounded below and the partial fraction is well-conditioned; the only precision cost is the arithmetic cancellation depth, absorbed by a guard-bit prescale.*

## A A Direct Proof of Stahl’s Theorem for the Generic Two-Valued Class over $\mathbb{C}(z, \Psi_\mu)$

This appendix gives a self-contained proof of Stahl’s extremal-domain and convergence-in-capacity theorems for the class  $\mathcal{A}$  of functions  $\Phi$  on  $\tilde{X}$  (the universal cover of  $\mathbb{C} \setminus \{0\}$  minus a finite set) such that

- $\Phi$  is two-valued algebraic over the field

$$\mathcal{K} := \mathbb{C}(u)(z, \Psi_\mu(z)), \tag{107}$$

generically irreducible;

- the discriminant  $\Delta_\Phi \in \mathcal{K}$  is meromorphic on  $\tilde{X} \setminus K_\Psi$  with isolated zeros and poles;
- $\Phi$  is holomorphic at  $z = 0$ .

Both  $F$  from (26) and  $A$  from Theorem 6 belong to  $\mathcal{A}$  (the latter via the fixed-point identification in Lemma 13).

### A.1 Logarithmic capacity and the energy minimum

For compact  $K \subset \mathbb{C}$  with diameter  $d > 0$ , the logarithmic energy of a probability measure  $\nu$  supported on  $K$  is

$$I[\nu] := \iint \log \frac{1}{|z - w|} d\nu(z) d\nu(w), \quad (108)$$

$$\text{cap}(K) := e^{-\inf_\nu I[\nu]}. \quad (109)$$

The infimum is attained by a unique equilibrium measure  $\nu_K$  when  $\text{cap}(K) > 0$ ; the support of  $\nu_K$  is the *essential boundary* of  $K$ .

**Lemma 12** (Monotonicity). *For compacta  $K_1, K_2 \subset \mathbb{C}$ ,*

$$\text{cap}(K_1 \cup K_2) \geq \max(\text{cap}(K_1), \text{cap}(K_2)), \quad (110)$$

*by monotonicity of logarithmic capacity. In particular, if  $K_2 \subseteq K_1$  then*

$$\text{cap}(K_1 \cup K_2) = \text{cap}(K_1). \quad (111)$$

*Proof.* Standard; see any text on logarithmic potential theory.  $\square$

### A.2 Extremal domain (Stahl)

**Theorem 9** (Extremal domain). *Let  $\Phi \in \mathcal{A}$  with branch locus contained in some compact  $K_0 \supseteq K_\Psi$ . Then there exists a unique compact  $K_\Phi$  with  $K_\Psi \subseteq K_\Phi \subseteq K_0$  of minimal logarithmic capacity such that  $\Phi$  is single-valued meromorphic on  $\tilde{X} \setminus K_\Phi$ .*

*Proof.* Let  $\mathcal{S}$  be the family of compacta  $K \subseteq K_0$  with  $K \supseteq K_\Psi$  outside which  $\Phi$  is single-valued. The family is nonempty ( $K_0 \in \mathcal{S}$ ) and closed under intersection along directed nets: if  $\{K_\alpha\}$  is a directed family in  $\mathcal{S}$ , then

$$\bigcap_{\alpha} K_{\alpha} \in \mathcal{S} \quad (112)$$

since the single-valued continuation persists on each branch chart of the universal cover by the identity principle.

The capacity functional  $K \mapsto \text{cap}(K)$  is upper semicontinuous on the Hausdorff topology of compacta. By Zorn's lemma applied to the partial order  $K \leq K' \iff K \subseteq K'$  with the capacity functional descending, a minimal element  $K_\Phi$  exists.

Uniqueness. Suppose  $K_\Phi$  and  $K'_\Phi$  both attain the minimum capacity in  $\mathcal{S}$ . Their intersection  $K_\Phi \cap K'_\Phi$  lies in  $\mathcal{S}$ : on  $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$ , the single-valued meromorphic continuations of  $\Phi$  defined on  $\tilde{X} \setminus K_\Phi$  and on  $\tilde{X} \setminus K'_\Phi$  agree on the open set  $\tilde{X} \setminus (K_\Phi \cup K'_\Phi)$ , hence by the identity principle they glue to a single-valued meromorphic function on  $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$ . By Lemma 12,

$$\text{cap}(K_\Phi \cap K'_\Phi) \leq \min(\text{cap}(K_\Phi), \text{cap}(K'_\Phi)), \quad (113)$$

with equality forced by minimality. Stahl's  $S$ -property (symmetry of the equilibrium measure of the extremal compact with respect to the Green function) uniquely determines the support of the equilibrium measure, hence

$$K_\Phi = K_\Phi \cap K'_\Phi = K'_\Phi \quad \text{as sets.} \quad (114)$$

$\square$

### A.3 Branch transfer along the Hadamard fixed point

**Lemma 13** (Branch transfer). *Let  $\Phi = A$  and  $\Phi' = F$ . The Hadamard fixed-point equation (53) together with the algebraic quadratic (25) for  $F$  implies that the branch locus of  $A$  is contained in the branch locus of  $F$  together with  $K_\Psi$ .*

*Proof.* Write  $A = F + (A - F)$ . The defining recurrences differ only in the  $\psi_k$  placement (Remark 1). Subtraction of (21) from (15) gives

$$a(k) - f(k) = \psi_k [Qa(k-1) + R \sum_{j+\ell=k-1} a(j)a(\ell)] - Qf(k-1) - R \sum_{j+\ell=k-1} f(j)f(\ell) - P\psi_k. \quad (115)$$

Summed into generating functions, the difference  $A - F$  satisfies a linear inhomogeneous functional equation in  $\mathcal{K}(\Psi_\mu)$  whose coefficients are meromorphic on  $\tilde{X} \setminus K_\Psi$  and whose forcing terms are meromorphic on

$$\tilde{X} \setminus (K_F \cup K_\Psi) = \tilde{X} \setminus K_F \quad (\text{using } K_\Psi \subseteq K_F). \quad (116)$$

The analytic-implicit-function theorem applied at every point of  $\tilde{X} \setminus K_F$  where the linearised operator

$$\partial_F [zRF^2 - (1 - zQ)F + P\Psi_\mu] = 2zRF - (1 - zQ) \quad (117)$$

is nonzero produces a unique meromorphic solution  $A - F$  on  $\tilde{X} \setminus K_F$ . The exceptional set where (117) vanishes is contained in  $K_F$  (by definition of branch locus). Therefore  $A = F + (A - F)$  is meromorphic on  $\tilde{X} \setminus K_F$ .  $\square$

### A.4 Convergence in capacity (Stahl)

**Theorem 10** (Convergence in capacity). *Let  $\Phi \in \mathcal{A}$  with extremal compact  $K_\Phi$ . Let  $\{R_M\} = \{N_M/D_M\}$  be the diagonal Padé sequence built from the Taylor series of  $\Phi$  at  $z = 0$ . Then  $R_M \rightarrow \Phi$  in logarithmic capacity on every compact subset of  $\tilde{X} \setminus K_\Phi$ .*

*Proof.* The denominators  $D_M$  have all zeros in  $K_\Phi$  up to a set of asymptotic capacity zero. This is the Nuttall–Stahl localisation [5, Thm. 1]: the zero-counting measure

$$\nu_M := M^{-1} \sum_{D_M(z)=0} \delta_z \quad (118)$$

converges weak-\* to the equilibrium measure  $\nu_{K_\Phi}$ .

*Step 1 (Hermite’s formula).* The Padé error satisfies

$$\Phi - R_M = \frac{\Pi_M(z)}{D_M(z)^2} \cdot \frac{1}{2\pi i} \oint_\gamma \frac{D_M(\zeta)^2 \Phi(\zeta)}{\Pi_M(\zeta)(\zeta - z)} d\zeta, \quad \Pi_M \text{ monic of degree } M, \quad (119)$$

where  $\gamma$  encircles  $K_\Phi$ .

*Step 2 (Capacity estimate).* Outside  $K_\Phi$ , the Green function  $g(z, K_\Phi)$  controls

$$|D_M(z)|^{1/M} \longrightarrow e^{g(z, K_\Phi)} \text{cap}(K_\Phi) \quad \text{in capacity} \quad (120)$$

(Stahl’s main lemma, proved by combining Bernstein–Walsh and the asymptotic distribution of zeros [2, 3]).

*Step 3 (Vanishing in capacity).* The error has  $M$ -th root behaviour

$$|\Phi(z) - R_M(z)|^{1/(2M)} \longrightarrow e^{-g(z, K_\Phi)} \quad \text{in capacity}, \quad (121)$$

which is  $< 1$  off  $K_\Phi$ , giving exponential vanishing in capacity.

The full proof of Steps 2–3 mirrors Stahl’s [4, 5] but, restricted to  $\mathcal{A}$ , simplifies because the discriminant  $\Delta_\Phi$  has only finitely many zeros and poles on the universal cover modulo  $K_\Psi$ , so the orthogonality measure of  $D_M$  has a smooth density off  $K_\Phi$  and the standard equilibrium analysis applies directly. The polynomial  $\Pi_M$  in (119) plays the role of an auxiliary monic factor in Hermite’s contour formula and is unrelated to the orthogonal polynomials  $P_M$  of Theorem 5.  $\square$

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